

Lab 1 Network Programming

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Server

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pramitha@pramitha-VirtualBox: ~/labsheet2
GNU nano 7.2 lab1_py_echo_server.py
import socket
import argparse

HOST = '0.0.0.0'

def server(port):
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    sock.bind((HOST, port))
    sock.listen(5)
    sock.settimeout(10) # <-- timeout added

    print("Server listening on", port)

    try:
        conn, addr = sock.accept()
        print("Connected:", addr)

        data = conn.recv(1024)
        while data:
            response = b"Name: Sandeepana Rupasingha | IT No: IT24102241"
            conn.sendall(response)
            data = conn.recv(1024)

        conn.close()

    except socket.timeout:
        print("No client connected within timeout period")

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--port', type=int, required=True)
    server(parser.parse_args().port)
```

```
pramitha@pramitha-VirtualBox: ~/labsheet2
pramitha@pramitha-VirtualBox: ~/labsheet2
pramitha@pramitha-VirtualBox:~/labsheet2$ nano lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$ python3 lab1_py_echo_server.py --port 5000
File "/home/pramitha/labsheet2/lab1_py_echo_server.py", line 5
    sock=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
    ^
IndentationError: expected an indented block after function definition on line 4
pramitha@pramitha-VirtualBox:~/labsheet2$ nano lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$ python3 lab1_py_echo_server.py --port 5000
Server listening on 5000
Connected: ('127.0.0.1', 43658)
pramitha@pramitha-VirtualBox:~/labsheet2$ nano lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$ python3 lab1_py_echo_server.py --port 5000
Server listening on 5000
No client connected within timeout period
pramitha@pramitha-VirtualBox:~/labsheet2$ python3 lab1_py_echo_server.py --port 5000
Server listening on 5000
No client connected within timeout period
pramitha@pramitha-VirtualBox:~/labsheet2$ python3 lab1_py_echo_server.py --port 5000
Server listening on 5000
Connected: ('127.0.0.1', 38872)
pramitha@pramitha-VirtualBox:~/labsheet2$ ls -l
total 24
-rwxrwxr-x 1 pramitha pramitha 16368 Feb 16 11:29 client
-rw-rw-r-- 1 pramitha pramitha 504 Feb 16 10:54 lab1_c_echo_client.c
-rw-rw-r-- 1 pramitha pramitha 871 Feb 16 11:25 lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$
```

Client

```
pramitha@pramitha-VirtualBox: ~/labsheet2
GNU nano 7.2 lab1_c_echo_client.c
/* lab1_c_echo_client.c */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
int main(int argc,char **argv){
    int sock=socket(AF_INET,SOCK_STREAM,0);
    struct sockaddr_in srv; srv.sin_family=AF_INET;
    srv.sin_port=htons(atoi(argv[2]));
    inet_pton(AF_INET,argv[1],&srv.sin_addr);
    connect(sock,(struct sockaddr*)&srv,sizeof(srv));
    char *msg="Test message";
    send(sock,msg,strlen(msg),0);
    char buf[128]; int n=recv(sock,buf,127,0);
    buf[n]=0; printf("Received:%s\n",buf);
}
```

```
pramitha@pramitha-VirtualBox: ~/labsheet2
Received:Test message
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
./client IT24102241
Segmentation fault (core dumped)
pramitha@pramitha-VirtualBox:~/labsheet2$ IT24102241
IT24102241: command not found
pramitha@pramitha-VirtualBox:~/labsheet2$ ./client IT24102241 5000
pramitha@pramitha-VirtualBox:~/labsheet2$ ./client Rupasingha IT24102241 5000
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
./client 127.0.0.1 5000
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
./client 127.0.0.1 5000
pramitha@pramitha-VirtualBox:~/labsheet2$ ls -l
total 24
-rwxrwxr-x 1 pramitha pramitha 16368 Feb 16 11:26 client
-rw-rw-r-- 1 pramitha pramitha 504 Feb 16 10:54 lab1_c_echo_client.c
-rw-rw-r-- 1 pramitha pramitha 871 Feb 16 11:25 lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
./client 127.0.0.1 5000
pramitha@pramitha-VirtualBox:~/labsheet2$ gcc lab1_c_echo_client.c -o client
./client 127.0.0.1 5000
Received:Name: Sandeepana Rupasingha | IT No: IT24102241
pramitha@pramitha-VirtualBox:~/labsheet2$ ls -l
total 24
-rwxrwxr-x 1 pramitha pramitha 16368 Feb 16 11:29 client
-rw-rw-r-- 1 pramitha pramitha 504 Feb 16 10:54 lab1_c_echo_client.c
-rw-rw-r-- 1 pramitha pramitha 871 Feb 16 11:25 lab1_py_echo_server.py
pramitha@pramitha-VirtualBox:~/labsheet2$
```

1) How does communication work even though server is in Python and client is in C?

Even though the server is written in Python and the client is written in C, the communication works without any issue because both programs use the same **TCP/IP protocol**.

TCP/IP is a standard network protocol handled by the operating system. It does not depend on the programming language. Both Python and C use the OS networking stack to:

- Create sockets
- Send data as bytes
- Receive data as bytes

When the client sends a message, it is converted into binary data (bytes). The server receives those bytes and sends them back. Since TCP/IP rules are followed by both programs, the communication works correctly.

So, the important point is:

Network protocols matter, not the programming language.

Function	Used By	Purpose
bind()	Server	Reserve IP & port
listen()	Server	Wait for clients
connect()	Client	Connect to server